Physical Features of India

Table of Contents

- Boundaries of India
- Border Lines Between India and Neighboring Countries
- Time Difference Between Gujarat & Arunachal Pradesh
- Coastal States of India
- Tropic of Cancer in India
- Physiographic Divisions Of India

Introduction

- The name "**India**" originates from the Indus River.
- It is the **seventh-largest** country in the world and the **second-largest in Asia**.
- India is situated in the Northeastern Hemisphere.
- Due to its diverse physiography, climate, natural vegetation, mineral resources, and human geography, India is often referred to as a "**Subcontinent.**"

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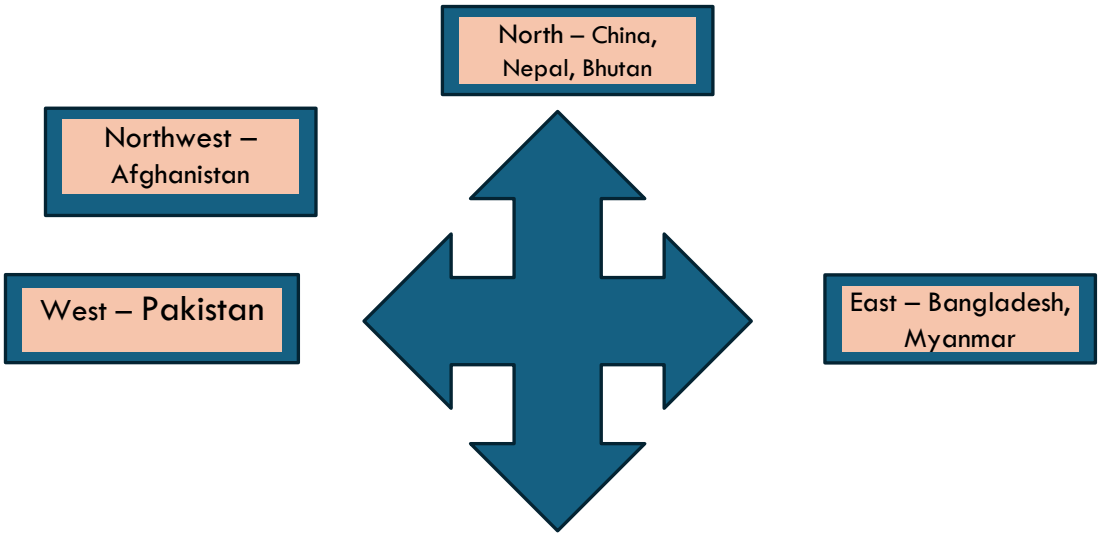
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Feature	Specification
Area	3,287,263 km² (~2.4% of world land area)
North–South Extent	3,214 km (Indira Col → Kanyakumari)
East–West Width	2,933 km (Rann of Kutch → Arunachal)
Latitudinal Span	8°4′ N to 37°6′ N (mainland)
Southernmost Point	**Indira Point,** ≈ 6°45′ N - in the Andaman and Nicobar Islands.
Southernmost point of mainland India	Cape Comorin (Kanyakumari)
Total Land Border	~15,200 km
Longest Border	**Bangladesh** – ~4,096 km
Shortest Border	**Afghanistan** – ~106 km
long coastline	6,100 km
Total coastline, including islands	7,516.6 km.
land borders	17 Indian states
coastal boundaries.	9 states
State Borders	**Uttar Pradesh** shares borders with the most states (8 states).
Indian states bordering only one state	1. Sikkim (borders West Bengal) 2. Meghalaya (borders Assam)

Boundaries of India



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Border Lines Between India and Neighboring Countries

Neighboring Country	Border Line Name	Border Length (km)
Pakistan	Radcliffe Line	3,323 km
Bangladesh	Burbach Line	4,096.7 km
China	McMahon Line	3,380 km
Bhutan	Indo-Bhutan Border	699 km
Afghanistan	Durand Line	106 km
Sri Lanka	Palk Strait	30 km
Myanmar	Indo-Burma Border	1,643 km
Nepal	Indo-Nepal Border	1,236 km

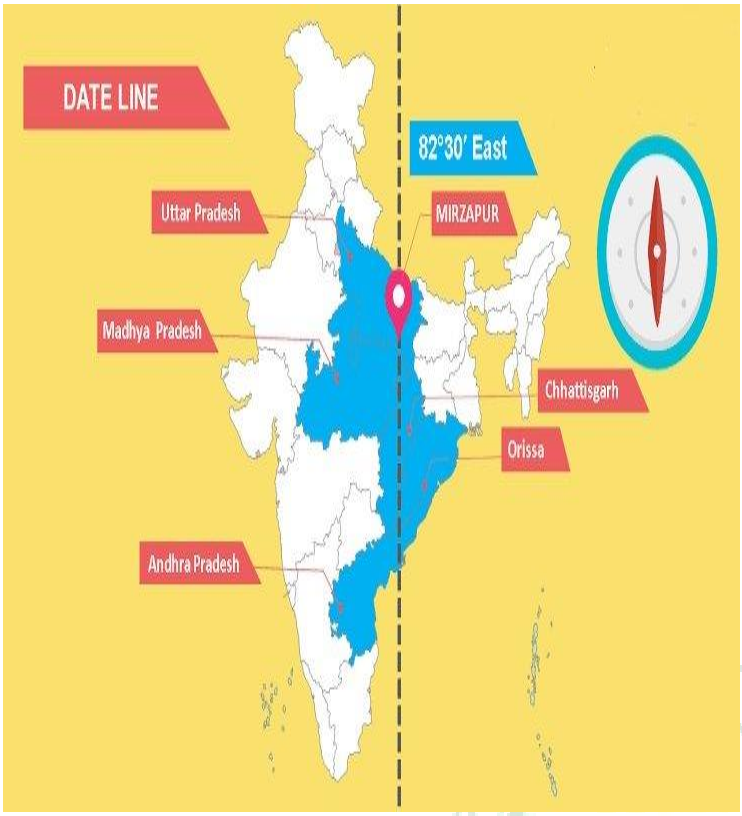
Physical and Climatic Characteristics

India is geographically separated from the rest of Asia by the Himalayan mountain range.

- The country's climate is predominantly influenced by tropical monsoons.
- The Himalayas act as a natural barrier, preventing cold temperate air masses from penetrating the region.
- Despite low winter night temperatures, clear skies and intense sunlight during the day contribute to significant temperature increases.

Time Difference Between Gujarat & Arunachal Pradesh

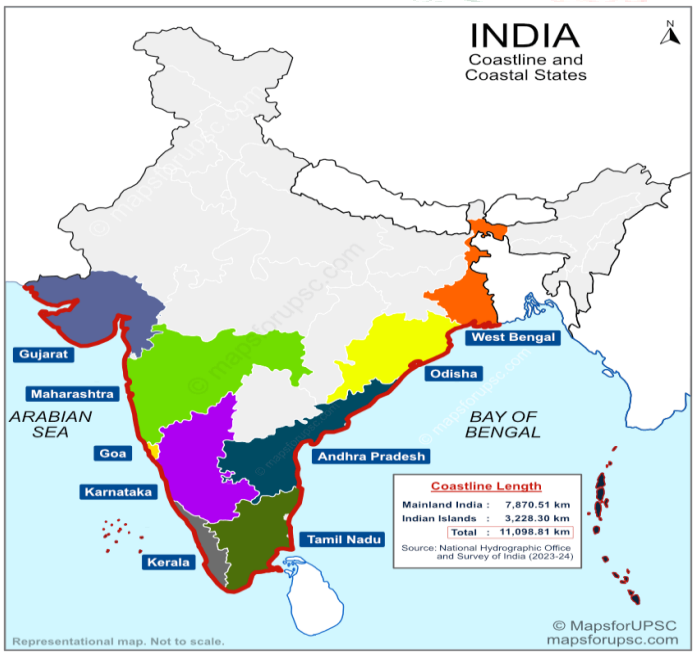
- **Longitude span:** ~29°18′ (≈30°) from Gujarat to Arunachal Pradesh.
- **Time rotation rate:** 1° longitude = 4 minutes.
- **Total difference:** 29.3° × 4 min ≈ 117 minutes (~1 h 57 min), rounded to 2 hours.
- **Effect:** Sun rises ~2 hours earlier in Arunachal Pradesh compared to Gujarat.



Indian Standard Meridian & IST

- **Standard Meridian:** 82°30′ E — passes near Mirzapur (Allahabad), Uttar Pradesh.
- **Time zone:** Entire country follows IST = UTC +5 h 30 min.
- **Why one zone:** India's 30° east–west spread would naturally give a ~2-hour solar time difference; using a single central meridian keeps things uniform

Coastal States of India



India has nine coastal states, namely:

1. Gujarat
2. Maharashtra
3. Goa
4. Karnataka
5. Kerala
6. Tamil Nadu
7. Andhra Pradesh
8. Odisha

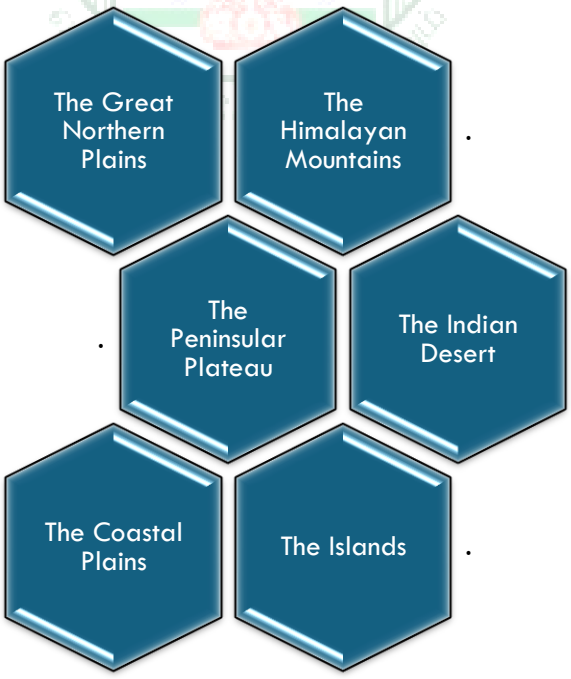
Tropic of Cancer in India

The Tropic of Cancer (23°30' N) passes through the middle of India, dividing the country into two halves.

- It crosses through eight states:
  - Gujarat – Jasdan
  - Rajasthan – Kalinjarh
  - Madhya Pradesh – Shajapur
  - Chhattisgarh – Sonhat
  - Jharkhand – Lohardaga
  - West Bengal – Krishnanagar
  - Tripura – Udaipur
  - Mizoram – Champhai

The Tropic of Cancer also passes through Jabalpur in Madhya Pradesh.

Physiographic Divisions of India

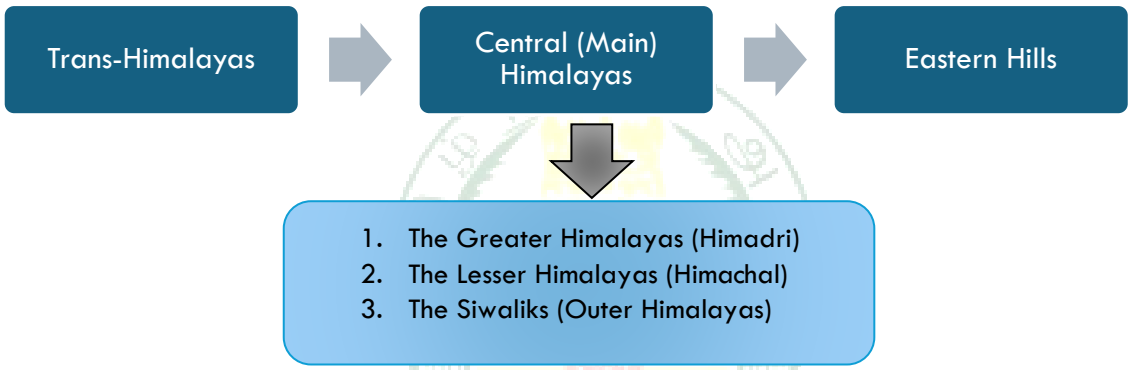


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The Himalayan Mountains

- Among the youngest and tallest fold mountains, the Himalayas were formed millions of years ago by the collision of the Indian and Eurasian plates—still rising ~5 mm/year
- **"Himalaya" means "Abode of Snow"** in Sanskrit
- They stretch in a **grand arc from the Pamir Knot—the "Roof of the World"** connecting to Central Asia—eastwards across ~2,500 km

The Northern Mountains act as a natural barrier and are divided into three sub-divisions:



Trans-Himalayas (Tibetan Himalayas):

- North of the main Himalayas, across Jammu & Kashmir and Tibet.
- Includes Zaskar, Ladakh, Kailash, and Karakoram ranges.
- Composed of ancient granites and marine fossils from the Tethys Sea

Central (Main) Himalayas – a three-tier structure:

- **Greater Himalayas (Himadri):** Continuous highest range (avg. 6,000 m), containing world's tallest peaks like Everest (8,849 m) and Kanchenjunga (8,586 m), with large glaciers
- **Lesser Himalayas (Himachal):** Mid-range (3,500–4,500 m), including Pir Panjal, Dhauladhar, Mahabharat; hosts popular hill stations (Shimla, Darjeeling)
- **Outer Himalayas (Siwaliks/Shiwaliks):** Foothills (600–1,500 m), 10–50 km wide, formed from alluvial debris; includes fertile Duns (west) and Duars (east) valleys

Prominent Peaks in the Himalayas



Greater Himalayas (Himadri)

- **Location & Profile:** Northmost, most continuous belt; avg. width ~25 km, elevation ~6,000 m (20,000 ft)
- **Climate:** Receives less rainfall than the middle and foothill ranges; blocks cold northern winds and draws heavy monsoon moisture on its southern slopes .
- **Features:**
- Rises the world's tallest peaks, including Mt. Everest (8,848 m) and Kanchenjunga (8,586 m)
- **Hosts major glaciers:** Gangotri, Yamunotri, Siachen (the world's longest non-polar glacier, ~76 km)

Lesser Himalayas (Himachal / Middle Himalayas)

- **Dimensions:** Up to ~80 km wide in places, averaging 3,500–4,500 m in elevation
- **Geology:** Composed of slate, limestone, quartzite, and crystalline rocks like granite and gneiss .
- Ranges & Tourism:
- Includes Pir Panjal, Dhauladhar, Mahabharat, Nag Tibba
- **Famous hill towns:** Shimla, Mussoorie, Nainital, Almora, Ranikhet, Darjeeling

Major Passes in the Lesser Himalayas

- Pir Panjal Pass -Jammu & Kashmir Jammu
- Banihal Pass - Jammu & Kashmir
- Rohtang Pass - Himachal Pradesh

Outer Himalayas (Siwaliks / Shiwaliks))

- Extent: Stretch from NW India to Assam, ~10–50 km wide, avg. elevation 900–1,100 m
- Composition: Formed from unconsolidated Tertiary sediments (sandstone, conglomerate, mudstone); characterized by anticlines and synclines
- Forests: Dense in the east, thinning westward; southern slopes in Himachal and Punjab nearly deforested
- Duns & Duars:
- Valleys such as Dehradun (~75 × 15–20 km), Kotli, Patli, etc. formed by river-blockage, lake formation, sediment fill, and eventual drainage
- "Duns" in the west; "Duars" (or Doon/Duar) in the east; ideal for agriculture and settlements

Purvanchal Hills (Eastern Hills)

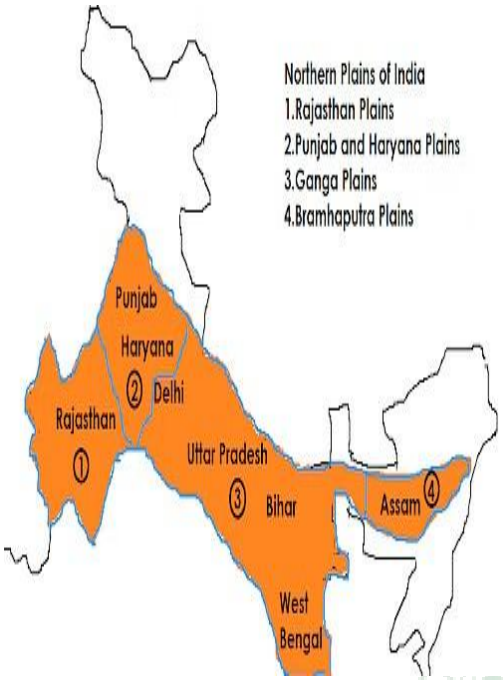
- These hills are eastern extensions of the Himalayas, mainly along the India-Myanmar border.
- Some extend inside India, forming important hill ranges.

Importance of the Himalayas

- **Monsoon Barrier:** Blocks Southwest Monsoon Winds, causing heavy rainfall in North India.
- **Natural Border:** Acts as a protective barrier for the Indian subcontinent.
- **Source of Major Rivers:** Home to perennial rivers such as Indus, Ganga, Brahmaputra, and their tributaries.
- **Tourism & Pilgrimage:** Houses famous hill stations and pilgrim centers like Amarnath, Kedarnath, Badrinath, and Vaishno Devi.
- **Climate Protection:** Prevents cold winds from Central Asia, protecting India from severe winters. fauna.

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The Great Northern Plains



1. The Rajasthan Plains

- Arid and semi-arid region.
- Home to salt lakes like Sambhar Lake and Pushkar Lake.

2. The Punjab-Haryana Plains

- Formed by the deposition of the Sutlej, Beas, and Ravi rivers.
- Known for highly fertile land and extensive agriculture.

3. The Ganga Plains

- Extends from the Yamuna River in the west to Bangladesh in the east.
- Covers about 3.75 lakh sq. km.
- Highly fertile, densely populated agricultural region.

4. The Brahmaputra Plains

- Covers about 56,275 sq. km.
- Features alluvial fans and marshy tracts due to seasonal floods.
- Rich biodiversity along the Brahmaputra River.



1. The Bhabar Plain

- Made of gravel and unassorted sediments from Himalayan rivers.
- High porosity causing small streams to disappear underground.
- Located along the foothills of the Himalayas, from Punjab to Assam.

2. The Tarai Tract

- Zone of excessive dampness, thick forests, and rich wildlife.
- Streams re-emerge from the Bhabar region, creating marshy conditions.
- Located south of the Bhabar belt, stretching from Uttarakhand to Assam.

3. The Bhangar Plains

- Older alluvial plains above the floodplain.
- Formed by ancient river deposits with calcareous concretions.
- Found in the upland areas of the Ganga and Yamuna plains.

4. The Khadar Plains

- New alluvial deposits found along riverbanks.
- Replenished annually with silt from floodwaters during the monsoon season.
- Highly fertile, suitable for intensive agriculture.
- Found along the Ganga, Yamuna, and Brahmaputra rivers.

5. The Deltaic Plains

- Triangle-shaped fertile land at the mouth of the Ganga and Brahmaputra rivers.
- Sundarban Delta is the largest and fastest-growing delta in the world.
- Soil types include old mud, new mud, and marshy land.
- Upland areas are called 'Chars' and marshy areas are called 'Bills'

The Peninsular Plateaus

- ✓ Lies to the south of the **Great Northern Plains.**
- ✓ Covers around 16 lakh sq. km (about half of India's total area).
- ✓ **Anaimudi (2,695 m)** is the **highest point** in the plateau.
- ✓ Located between the Narmada River and the Northern Great Plains.
- ✓ **Gurushikhar (1,722 m) is the highest peak.**
- ✓ The Chhota-Nagpur Plateau is the eastern part of the Central Highlands, located in the northeastern Indian Plateau.

Minor Plateaus in the Peninsular Plateau

Malwa Plateau

- ❖ Forms a triangular area between the Vindhyan Hills, Aravalli Range, and Madhya Bharat Pathar.
- ❖ Drained by rivers: Narmada, Tapti, Mahi (Arabian Sea), Chambal, Betwa (Bay of Bengal).
- ❖ Characterized by black soils, rolling terrain, and Chambal ravines.

Baghelkhand Plateau

- ❖ Located north of the Maikal Range, made of limestones, sandstones, and granite.
- ❖ Elevation ranges from 150 m to 1,200 m.
- ❖ Divides the Son and Mahanadi drainage systems.

Chotanagpur Plateau

- ❖ Located in northeastern India, mainly in Jharkhand.
- ❖ Features radial drainage (e.g., Damodar, Subarnrekaha).
- ❖ Coal fields like Gondwana are located here.

Meghalaya Plateau

- ❖ Includes Shillong Plateau and Karbi Anglong Plateau.
- ❖ Features Garo Hills, Khasi-Jaintia Hills, and Mikir Hills.
- ❖ Highest point: Shillong (1,961 m).

- **Maharashtra Plateau**
  - ❖ Located in Maharashtra, part of the Deccan Plateau.
  - ❖ Composed mainly of basaltic rocks (Deccan Traps).
  - ❖ Rolling plain appearance due to weathering.
- **Karnataka Plateau**
  - ❖ Also known as the Mysore Plateau.
  - ❖ Elevation: 600-900 m, dissected by rivers from the Western Ghats.
  - ❖ Highest peak: Mulangiri (1,913 m).
- **Marwar Plateau**
  - ❖ Divided into Malnad (hill country) and Maidan (rolling plains).
  - ❖ Located in eastern Rajasthan, west of the Aravallis.
  - ❖ Elevation: 250-500 m above sea level.
  - ❖ Composed of sandstone, shales, and limestones.
- **Telangana Plateau**
  - ❖ Elevation: 500-600 m, higher in the south.
  - ❖ Drained by rivers: Godavari, Krishna, Penneru.
- **Chhattisgarh Plain**
  - ❖ A saucer-shaped depression drained by the Mahanadi River.
  - ❖ Elevation: 250 m (east) to 330 m (west).

Deccan Plateau

- 🗺️ **Area**: ~500,000 km²
- 📐 **Borders**: Surrounded by hill ranges (Satpura–Vindhya north; Ghats east & west)
- ⛰️ **Elevation**: ~600 m avg (500 m north → 1,000 m south)
- 🌊 **Slope**: West → East, helping rivers flow eastward
- 🧱 **Division**: Rivers create smaller plateaus within

Feature	Western Ghats (Sahyadri)	Eastern Ghats (Poorvadri)
Continuity	Continuous, parallel to Arabian Sea coast	Disconnected hills—north, gap, reappear in south
Elevation Trend	Increases from north (~900 m) to south (~2,695 m)	Varies—higher in north and south, lower mid-section
Major Peaks	Anamudi (2,695 m), Doda Betta (2,637 m)	Mahendra Giri (1,501 m), Jindhagada (1,690 m), Biligiri Rangan (1,279 m)
Orientation	North–South along west coast	Southwest–Northeast along east coast
Key Geographical Gap	Palghat Gap divides Sahyadris	Interrupted between Godavari & Krishna, etc.

Aravalli Range

- Orientation & Extent: Stretches NE–SW for about 670 km, from Delhi through Haryana and Rajasthan, ending near Palanpur, Gujarat
- Antiquity: One of the oldest folded mountain ranges in the world—formed during the Proterozoic, now heavily eroded over millions of years

Elevation:

- Typically: 400–900 m across most of the range
- Mount Abu plateau: rises to ~1,158 m
- Guru Shikhar: highest peak at 1,722 m

Extent in Terrain:

- More continuous and prominent south of Ajmer (up to ~900 m)
- Less defined (discontinuous ridges) near Haryana and Delhi
- Actually extends beneath the alluvial plains up to Haridwar

Vindhya Range:

- Overlooks the Narmada Valley, rising as an escarpment along the northern edge of the Narmada-Son Trough.

❖ Extends east-west for over 1,200 km, from Jobat (Gujarat) to Sasaram (Bihar), running parallel to the Narmada Valley.
❖ General elevation: 300-650 m.
❖ Composed mainly of horizontally bedded ancient sedimentary rocks.
❖ Extends eastwards as the Barner and Kaimur hills.
❖ Acts as a watershed between the Ganga system and the river systems of South India.
❖ Rivers like Chambal, Betwa, and Ken rise within 30 km of the Narmada.

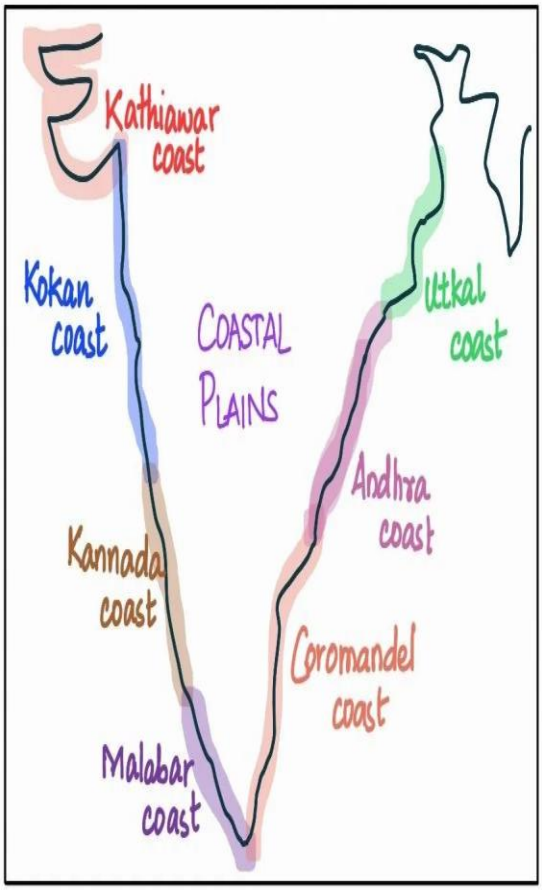
Satpura Range:

❖ Runs east-west, south of the Vindhyas, between the Narmada and Tapti rivers, parallel to them.
❖ Stretches for about 900 km.
❖ Parts of the range are folded and uplifted, forming a structural uplift or "horst."
❖ Dhupgarh (1,350 m) in the Mahadev Hills is the highest peak.
❖ Amarkantak (1,127 m) is another important peak.

Coastal Plains:

India's coastal plains

Feature	Eastern Coastal Plains	Western Coastal Plains
Width	Wider (35–200 km)	Narrower (10–65 km)
Delta formation	Large delta systems (Mahanadi, Godavari, etc.)	Smaller rivers, estuaries
Rainfall pattern	Receives NE monsoon (esp. south)	Receives SW monsoon (heavy)
Coastal type	Mostly emergent	Mostly submergent, some emergent
Ecology & economy	Rich in agriculture, fisheries, biodiversity	Known for ports, fishing, backwaters, minerals



Eastern Coastal Plains

- ✓ **Utkal Coast** (Chilika Lake → Kolleru Lake) – Heavy rainfall; grows rice, coconut, banana; features Mahanadi delta and Chilika Lagoon
- ✓ **Andhra Coast** (Kolleru → Pulicat) – Fertile plains from Godavari–Krishna basins; includes Pulicat Lagoon
- ✓ **Coromandel Coast** (Pulicat → Kanyakumari) – Dry summers, wet winters via northeast monsoon

Western Coastal Plains

- ✓ **Kachchh & Kathiawar Coast** – Formerly gulf regions; includes Great & Little Rann; seasonal flooding
- ✓ **Konkan Coast** (Daman → Goa) – Rice and cashew farming, heavy SW monsoon rainfall
- ✓ **Kanara Coast** (Mormugao → Mangalore) – Rich in iron ore; moderate width
- ✓ **Malabar Coast** (Mangalore → Kanyakumari) – Known for lagoons and backwaters (e.g., Vembanad), major tourist and navigation regions

The Islands

Major Island Groups in India

- ❖ Andaman and Nicobar Islands
- ❖ Lakshadweep Islands

Andaman and Nicobar Islands:

- ✓ Volcanic origin, located in the Bay of Bengal.
- ✓ It Comprises 572 islands.
- ✓ Area: 8,249 sq. km.
- ✓ Capital: Port Blair.
- ✓ **Divided into two groups:**
  - ○ Andaman (North): 25 inhabited islands.
  - ○ Nicobar (South): 13 inhabited islands.

- ✓ Car Nicobar: Northernmost island.
- ✓ Geologically, islands consist of tertiary sandstone, limestone, and shale resting on basic and ultrabasic volcanoes.
- ✓ Barren Island: India's only active volcano.
- ✓ Narcondam Island: An extinct/dormant volcano.
- ✓ Coral reefs and dense forests cover many of the islands.
- ✓ Saddle Peak (737 m) in North Andaman is the highest peak.
- ✓ Ten Degree Channel separates Andaman from Nicobar.
- ✓ Indira Point: Southernmost tip, part of Nicobar.

Lakshadweep Islands:

- ✓ Located in the Arabian Sea, consisting of:
  - o Amindivi Islands.
  - o Laccadive Islands (including Kavaratti).
  - o Minicoy.
- ✓ Known collectively as Lakshadweep (means "one lakh islands").
- ✓ Area: 32 sq. km.
- ✓ Capital: Kavaratti.
- ✓ 27 islands, 11 inhabited.
- ✓ Separated from the Maldive Islands by the Eight Degree Channel.
- ✓ Pitt Island: Uninhabited, has a bird sanctuary.
- ✓ Formerly divided into Laccadive, Minicoy, and Amindivi.
- ✓ Renamed Lakshadweep in 1973.